

13.2 Cholesky Decomposition



Core skills you will master in this section:

- ▶ Cholesky decomposition: applies only to positive definite matrices.
- ▶ Geometric perspective on Cholesky decomposition: "scaling → shear → shear → scaling".
- ▶ Another geometric perspective on Cholesky decomposition: "shear → scaling → scaling → shear".
- ▶ LDL decomposition: applies to symmetric matrices.
- ▶ Geometric perspective on LDL decomposition: "shear → scaling → shear".
- ▶ Both methods can be used to generate random numbers with a specified correlation.

This section introduces two special matrix decompositions: Cholesky decomposition and LDL decomposition. Cholesky decomposition applies only to positive definite matrices, while LDL decomposition applies to symmetric matrices.

Cholesky Decomposition

As shown in Figure 1, Cholesky decomposition decomposes matrix A into

$$A = R^T R$$

where R is upper triangular, and R^T is lower triangular.

⚠ Note that only a positive definite matrix admits a Cholesky decomposition.

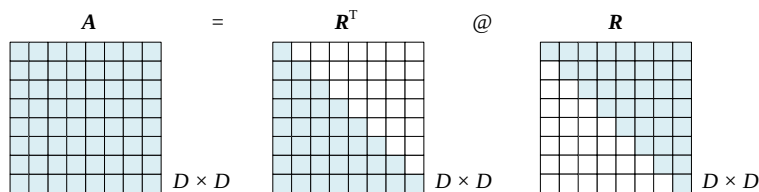


Figure 1. Cholesky Decomposition of a Positive Definite Matrix

A Geometric View of Cholesky Decomposition

As an example, given the following matrix A

$$A = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix}$$

We already performed eigenvalue decomposition on this matrix earlier in this book, and we know that A 's eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = 1/2$. The square matrix A is positive definite, so it admits a Cholesky decomposition.

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Comments and corrections are welcome. The book's dedicated email address: jiang.visualize.ml@gmail.com

Performing a Cholesky decomposition on $\mathbf{A}$ gives

$$\mathbf{A} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{5}/2 & 0 \\ -3\sqrt{5}/10 & 2\sqrt{5}/5 \end{bmatrix}}_{\mathbf{R}^T} @ \underbrace{\begin{bmatrix} \sqrt{5}/2 & -3\sqrt{5}/10 \\ 0 & 2\sqrt{5}/5 \end{bmatrix}}_{\mathbf{R}}$$

As shown in Figure 2, $\mathbf{A}\mathbf{x}$ can be written as $\mathbf{R}^T \mathbf{R}\mathbf{x}$, which means $\mathbf{R}$ first applies a linear transformation to $\mathbf{x}$, and then $\mathbf{R}^T$ acts.

You may have noticed that the geometric operation corresponding to $\mathbf{R}$ contains both a shear component and a scaling component; let's break this down below.

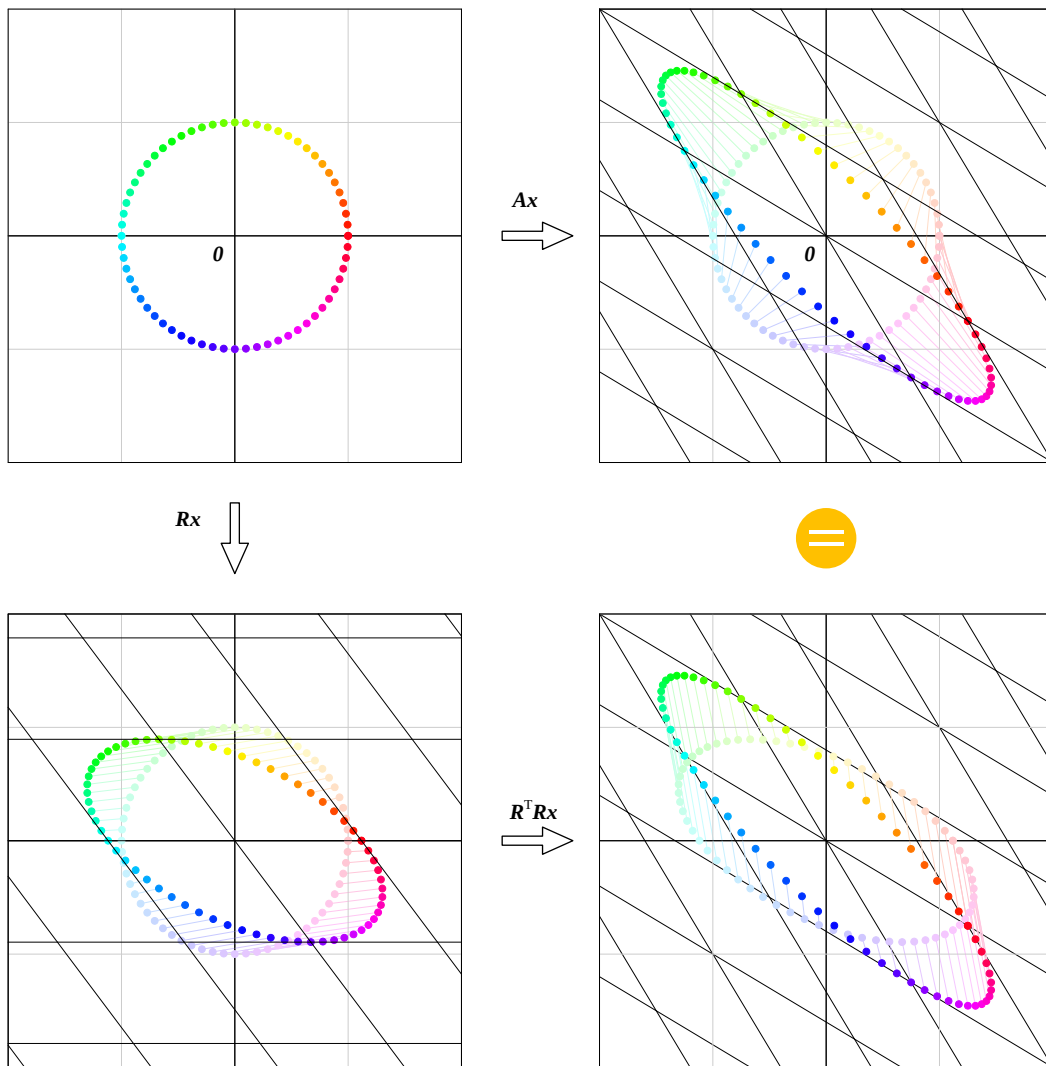


Figure 2. The Linear Transformation Corresponding to the Cholesky Decomposition of Matrix $\mathbf{A}$

Scaling $\rightarrow$ Shear $\rightarrow$ Shear $\rightarrow$ Scaling

In the equation above, $\mathbf{R}^T$ can be written as

$$\mathbf{R}^T = \begin{bmatrix} \sqrt{5}/2 & 0 \\ -3\sqrt{5}/10 & 2\sqrt{5}/5 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{5}/2 & 0 \\ 0 & 2\sqrt{5}/5 \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} 1 & 0 \\ -3/4 & 1 \end{bmatrix}}_{\mathbf{K}}$$

That is, $\mathbf{R}^T$'s corresponding geometric transformation can be understood as "shear $\rightarrow$ scaling". $\mathbf{K}$ shears along the vertical axis.

$\mathbf{R}$ can be written as

$$\mathbf{R} = \begin{bmatrix} \sqrt{5}/2 & -3\sqrt{5}/10 \\ 0 & 2\sqrt{5}/5 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & -3/4 \\ 0 & 1 \end{bmatrix}}_{\mathbf{K}^T} \underbrace{\begin{bmatrix} \sqrt{5}/2 & 0 \\ 0 & 2\sqrt{5}/5 \end{bmatrix}}_{\mathbf{S}}$$

$\mathbf{K}^T$ shears along the horizontal axis.

$\mathbf{R}$ can be understood as "scaling $\rightarrow$ shear".

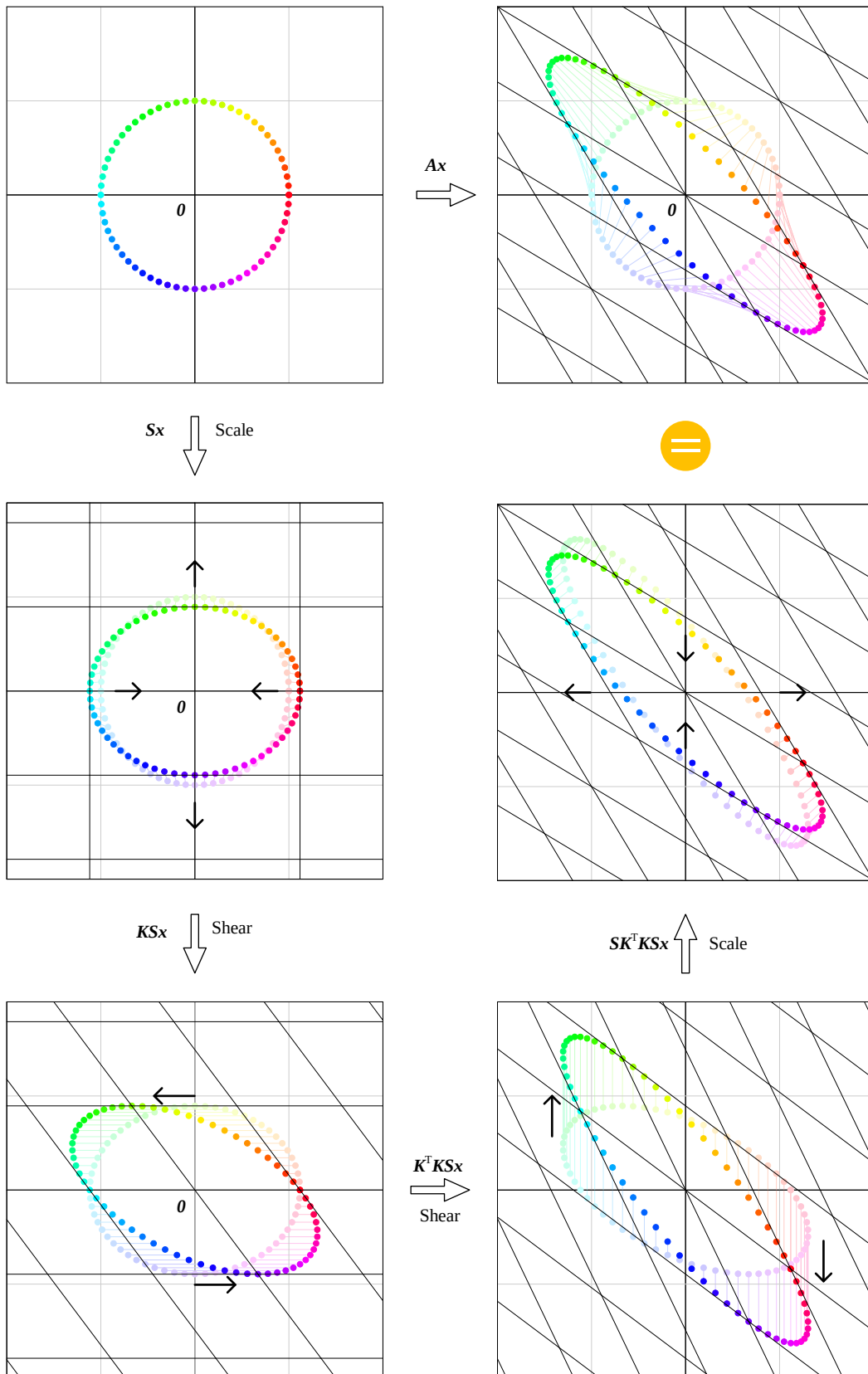
Thus, $\mathbf{A}$ can be written as

$$\mathbf{A} = \mathbf{R}^T \mathbf{R} = \mathbf{S} \mathbf{K} (\mathbf{S} \mathbf{K})^T = \mathbf{S} \mathbf{K} \mathbf{K}^T \mathbf{S}^T = \mathbf{S} \mathbf{K} \mathbf{K}^T \mathbf{S}$$

As shown in Figure 3, the linear transformation of $\mathbf{A}$ can thus be understood as "scaling ($\mathbf{S}$) $\rightarrow$ shear ($\mathbf{K}^T$) $\rightarrow$ shear ($\mathbf{K}$) $\rightarrow$ scaling ($\mathbf{S}$)".



We covered this in Chapter 8, Section 3 of this book — please review it if needed.

Figure 3. The Cholesky Decomposition of Matrix A Corresponds to "Scaling → Shear → Shear → Scaling"

Shear → Scaling → Scaling → Shear

In the equation above, $\mathbf{R}^T$ can also be written as

$$\mathbf{R}^T = \begin{bmatrix} \sqrt{5}/2 & 0 \\ -3\sqrt{5}/10 & 2\sqrt{5}/5 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ -3/5 & 1 \end{bmatrix}}_{\mathbf{L}} \underbrace{\begin{bmatrix} \sqrt{5}/2 & 0 \\ 0 & 2\sqrt{5}/5 \end{bmatrix}}_{\mathbf{S}}$$

That is, $\mathbf{R}^T$'s corresponding geometric transformation can be understood as "scaling $\rightarrow$ shear". $\mathbf{L}$ shears along the vertical axis.

$\mathbf{R}$ can be written as

$$\mathbf{R} = \begin{bmatrix} \sqrt{5}/2 & -3\sqrt{5}/10 \\ 0 & 2\sqrt{5}/5 \end{bmatrix} = \underbrace{\begin{bmatrix} \sqrt{5}/2 & 0 \\ 0 & 2\sqrt{5}/5 \end{bmatrix}}_{\mathbf{S}} \underbrace{\begin{bmatrix} 1 & -3/5 \\ 0 & 1 \end{bmatrix}}_{\mathbf{L}^T}$$

$\mathbf{L}^T$ shears along the horizontal axis.

Thus, $\mathbf{A}$ can be written as

$$\mathbf{A} = \mathbf{R}^T \mathbf{R} = \mathbf{L} \mathbf{S} (\mathbf{L} \mathbf{S})^T = \mathbf{L} \mathbf{S} \mathbf{S}^T \mathbf{L}^T = \mathbf{L} \mathbf{S} \mathbf{S} \mathbf{L}^T$$

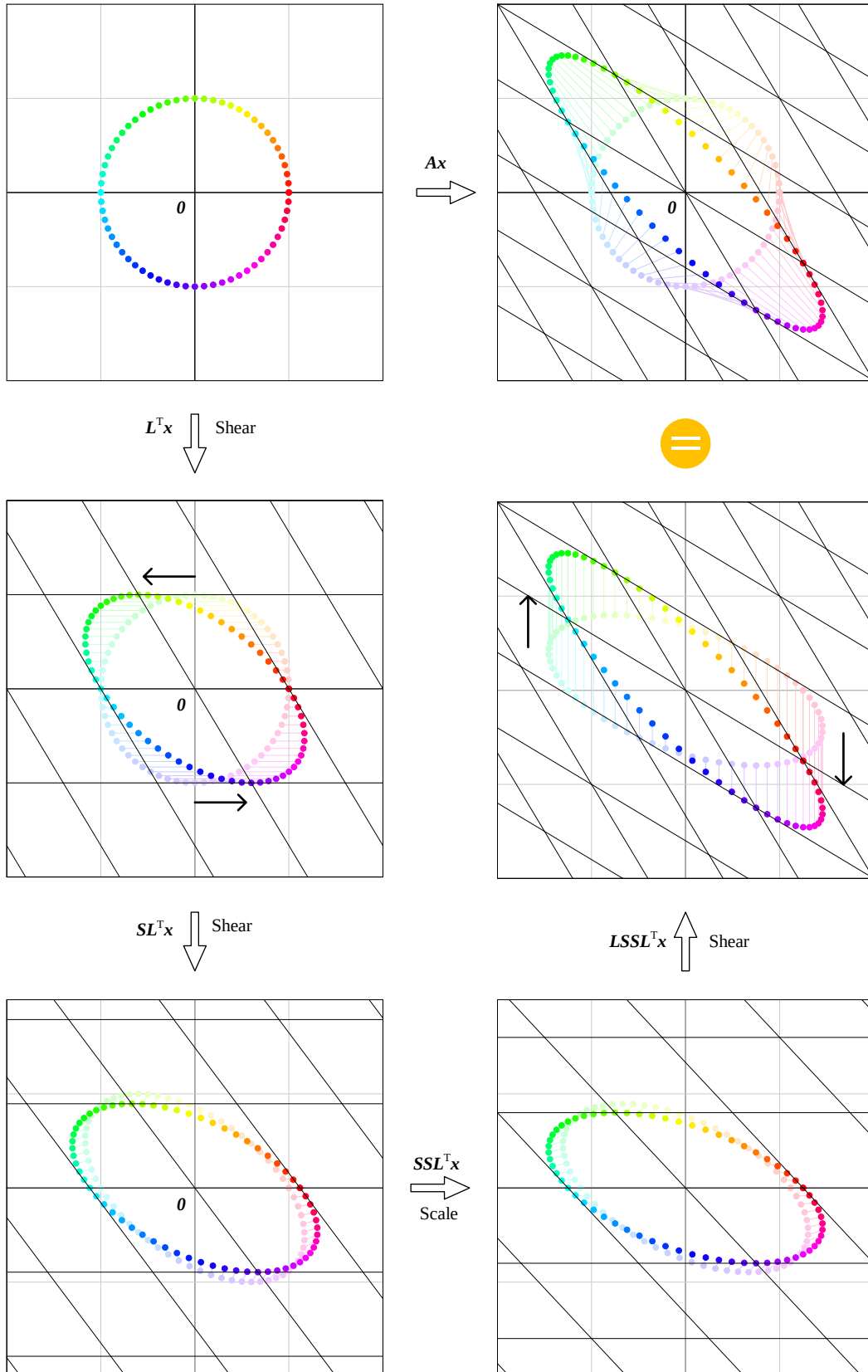
As shown in Figure 4, $\mathbf{A}$'s linear transformation can thus be understood as "shear ($\mathbf{L}^T$) $\rightarrow$ scaling ($\mathbf{S}$) $\rightarrow$ scaling ($\mathbf{S}$) $\rightarrow$ shear ($\mathbf{L}$)".

Combining the two $\mathbf{S}$ s in the equation above gives

$$\mathbf{A} = \mathbf{L} \mathbf{S}^2 \mathbf{L}^T$$

$\mathbf{A}$'s linear transformation can be understood as "shear ($\mathbf{L}^T$) $\rightarrow$ scaling ($\mathbf{S}^2$) $\rightarrow$ shear ($\mathbf{L}$)".

This actually leads us to the second matrix decomposition in this section — LDL decomposition.

Figure 4. The Cholesky Decomposition of Matrix A Corresponds to "Shear $\rightarrow$ Scaling $\rightarrow$ Scaling $\rightarrow$ Shear"

LDL Decomposition

As shown in Figure 5, Cholesky decomposition can be further generalized to LDL decomposition (LDL decomposition, LDL^T decomposition):

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$$\mathbf{A} = \mathbf{L}\mathbf{D}\mathbf{L}^T = \mathbf{L}\mathbf{D}^{1/2}(\mathbf{D}^{1/2})^T\mathbf{L}^T = \mathbf{L}\mathbf{D}^{1/2}(\mathbf{L}\mathbf{D}^{1/2})^T$$

where $\mathbf{L}$ is lower triangular with all diagonal elements equal to 1, and $\mathbf{D}$ is a diagonal matrix that performs the scaling.

From a geometric perspective, $\mathbf{L}$'s role is simply "shear". In other words, matrix $\mathbf{A}$ is decomposed into "shear $\rightarrow$ scaling $\rightarrow$ shear".

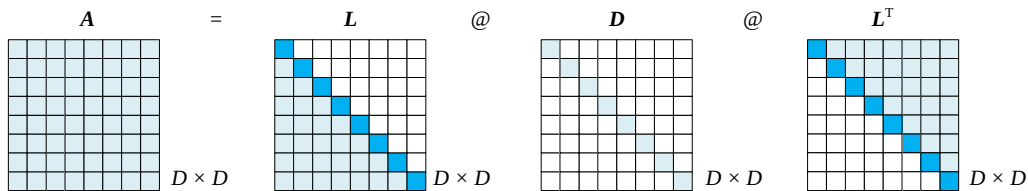


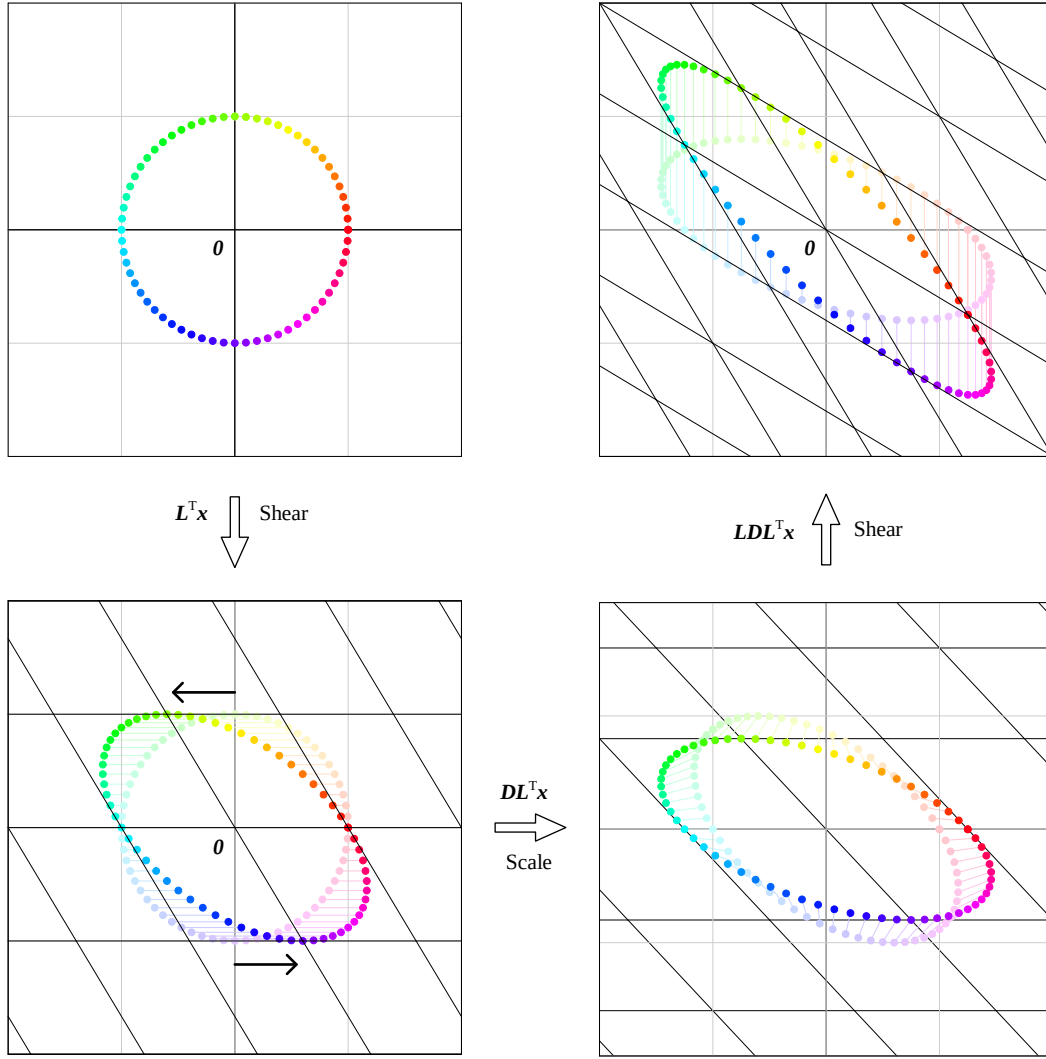
Figure 5. Illustration of the Matrix Operations in LDL Decomposition

Performing an LDL decomposition on matrix $\mathbf{A}$ from the equation above gives

$$\mathbf{A} = \begin{bmatrix} 5/4 & -3/4 \\ -3/4 & 5/4 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 \\ -3/5 & 1 \end{bmatrix}}_{\mathbf{L}} \underbrace{\begin{bmatrix} 5/4 & 0 \\ 0 & 4/5 \end{bmatrix}}_{\mathbf{D}} \underbrace{\begin{bmatrix} 1 & -3/5 \\ 0 & 1 \end{bmatrix}}_{\mathbf{L}^T}$$

The corresponding geometric operation is shown in Figure 6.

Therefore, Cholesky decomposition can be viewed as a special case of LDL decomposition, obtained by further taking the square root of the $\mathbf{D}$ from LDL decomposition.

Figure 6. The LDL Decomposition of Matrix A Corresponds to "Shear $\rightarrow$ Scaling $\rightarrow$ Shear"

Generating Random Numbers with a Given Correlation

Figure 7 shows random numbers whose centroid is at the origin, with covariance matrix equal to the identity matrix; writing the scattered data points in the figure as a data matrix gives Z .



Note that Z 's row vectors represent the scattered points in Figure 7.

The data matrix Z 's centroid being located at the origin means

$$\mu_z = \frac{Z^T \mathbf{1}}{n} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

where n is the number of sample data points.

The data matrix Z 's covariance matrix is then

$$\Sigma_z = \frac{Z^T Z}{n-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

Since the data matrix Z 's centroid is already the zero vector, the equation above involves no centering (mean removal).

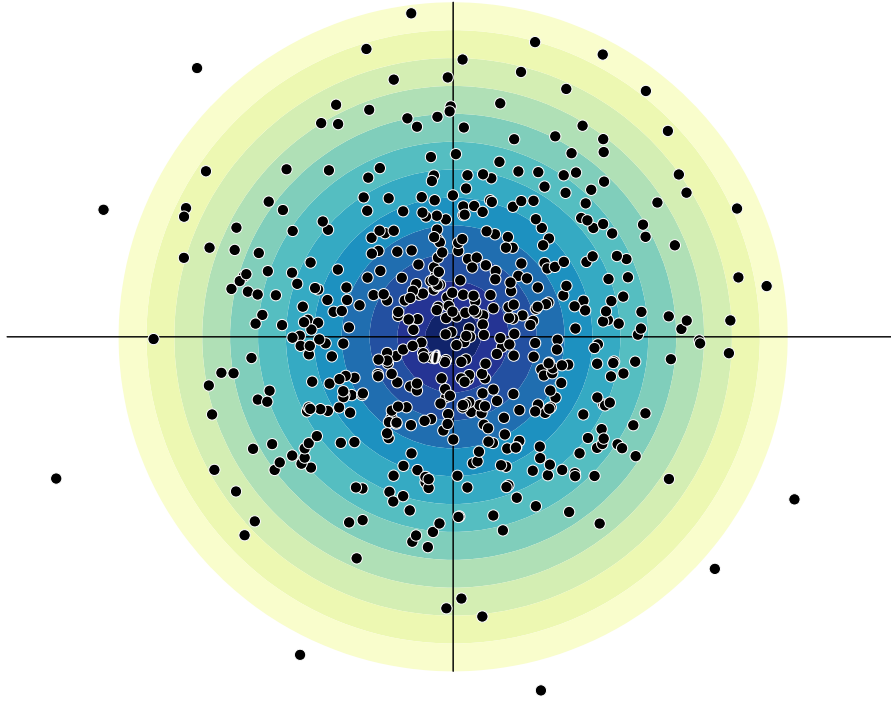


Figure 7. Data Matrix (Centroid at the Origin, Covariance Matrix Equal to the Identity Matrix) $\mathbf{Z}$

And Figure 8 shows random numbers whose centroid is located at $[4, 3]^T$, and whose covariance matrix is $[1, 1/2; 1/2, 1]$; writing the scattered data points in the figure as a data matrix gives $\mathbf{X}$.

The data matrix $\mathbf{X}$'s centroid being located at $[4, 3]^T$ means

$$\boldsymbol{\mu}_x = \frac{\mathbf{X}^T \mathbf{1}}{n} = \begin{bmatrix} 4 \\ 3 \end{bmatrix}$$

The data matrix $\mathbf{X}$'s covariance matrix is then

$$\boldsymbol{\Sigma}_x = \frac{\mathbf{X}_c^T \mathbf{X}_c}{n-1} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

where $\mathbf{X}_c$ is the centering of $\mathbf{X}$, that is

$$\mathbf{X}_c = \mathbf{X} - \boldsymbol{\mu}_x^T$$

Geometrically, the equation above corresponds to a translation.



Data centering was covered in Chapter 12, Section 1 of this book – please review it if needed.

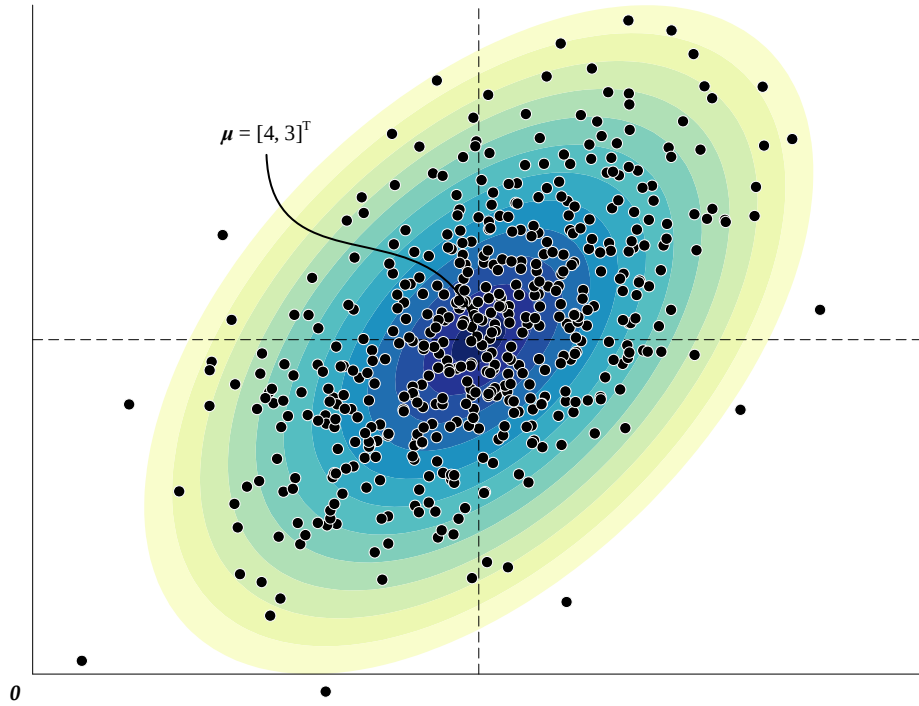


Figure 8. Data Matrix with Centroid at $[4, 3]^T$ and Covariance Matrix $[1, 1/2; 1/2, 1]$, Denoted $\mathbf{X}$

Next, we show how Cholesky decomposition and translation let us convert between $\mathbf{Z}$ and $\mathbf{X}$.

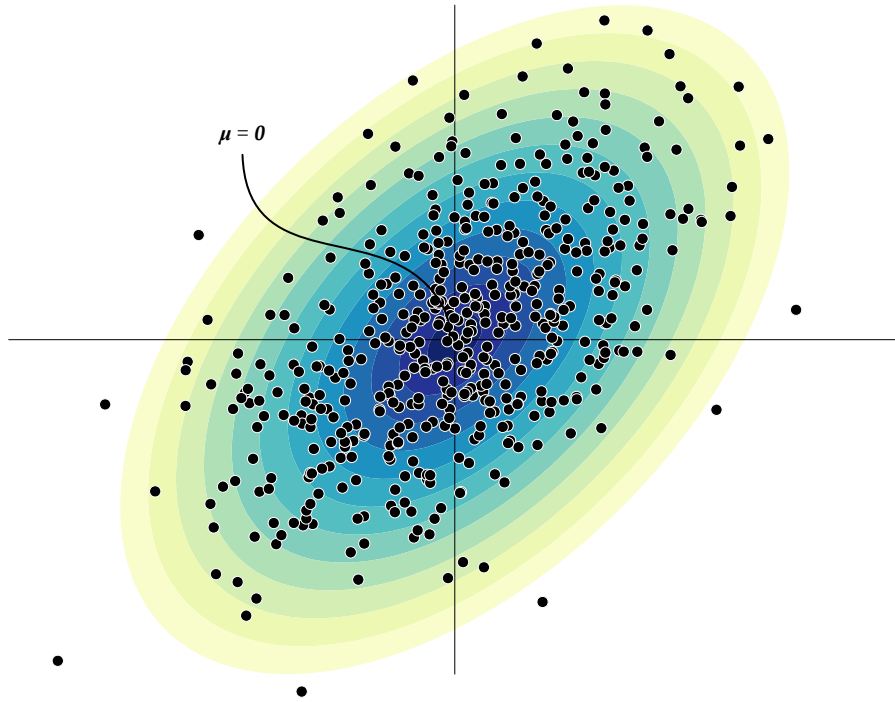
First, we perform a Cholesky decomposition on $\Sigma_{\mathbf{X}}$ from the equation above

$$\Sigma_{\mathbf{X}} = \mathbf{R}^T \mathbf{R} = \begin{bmatrix} 1 & 0 \\ 1/2 & \sqrt{3}/2 \end{bmatrix} \begin{bmatrix} 1 & 1/2 \\ 0 & \sqrt{3}/2 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

$\mathbf{X}$ can be written as

$$\mathbf{X} = \mathbf{Z}\mathbf{R} + \mu_{\mathbf{X}}^T$$

$\mathbf{Z}$ (Figure 7) transformed by $\mathbf{R}$ gives Figure 9, with the data's center still at the origin; then applying the broadcasting rule to translate the data gives Figure 8.

Figure 9. Data Matrix $\mathbf{Z} @ \mathbf{R}$

Verifying $\mathbf{X}$'s covariance matrix

$$\Sigma_{\mathbf{X}} = \frac{(\mathbf{X} - \mu_{\mathbf{X}}^T)^T (\mathbf{X} - \mu_{\mathbf{X}}^T)}{n-1} = \frac{\mathbf{R}^T \mathbf{Z}^T \mathbf{Z} \mathbf{R}}{n-1} = \mathbf{R}^T \frac{\mathbf{Z}^T \mathbf{Z}}{n-1} \mathbf{R} = \mathbf{R}^T \mathbf{R}$$

? As introduced earlier in this section, $\mathbf{R}$ can be further decomposed as "shear $\rightarrow$ scaling" or "scaling $\rightarrow$ shear" – please write out the corresponding matrix decomposition.

Looking at the equation above in reverse, the following operation shows that $\mathbf{X}$ can also become $\mathbf{Z}$

$$\mathbf{Z} = (\mathbf{X} - \mu_{\mathbf{X}}^T) \mathbf{R}^{-1}$$

Note that in the equation above, $\mathbf{R}$ is invertible; this means $\mathbf{R}$ has full rank. In other words, $\Sigma_{\mathbf{X}}$ is positive definite.

Clearly, the covariance matrix $\Sigma_{\mathbf{X}}$ is a symmetric matrix, so we can perform spectral decomposition on it. Since Cholesky decomposition can serve as a bridge between $\mathbf{X}$ and $\mathbf{Z}$, spectral decomposition can too.

Performing spectral decomposition on the covariance matrix $\Sigma_{\mathbf{X}}$

$$\Sigma_{\mathbf{X}} = \mathbf{V} \mathbf{\Lambda} \mathbf{V}^T = \begin{bmatrix} \sqrt{2}/2 & -\sqrt{2}/2 \\ \sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} \begin{bmatrix} 3/2 & 0 \\ 0 & 1/2 \end{bmatrix} \begin{bmatrix} \sqrt{2}/2 & \sqrt{2}/2 \\ -\sqrt{2}/2 & \sqrt{2}/2 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

Since the eigenvalues are all nonnegative, we can rewrite the equation above as

$$\Sigma_{\mathbf{X}} = \left(\mathbf{V} \mathbf{\Lambda}^{\frac{1}{2}} \right) \left(\mathbf{V} \mathbf{\Lambda}^{\frac{1}{2}} \right)^T = \begin{bmatrix} \sqrt{3}/2 & -1/2 \\ \sqrt{3}/2 & 1/2 \end{bmatrix} \begin{bmatrix} \sqrt{3}/2 & \sqrt{3}/2 \\ -1/2 & 1/2 \end{bmatrix} = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1 \end{bmatrix}$$

That is, the relationship between $\mathbf{X}$ and $\mathbf{Z}$ can also be written as

$$\mathbf{X} = \mathbf{Z} \left(\mathbf{V} \mathbf{A}^{\frac{1}{2}} \right)^T + \boldsymbol{\mu}_X^T = \mathbf{Z} \mathbf{A}^{\frac{1}{2}} \mathbf{V}^T + \boldsymbol{\mu}_X^T$$

The step-by-step geometric operations in the equation above are "scaling $\rightarrow$ rotation $\rightarrow$ translation".

Looking just at the scaling step, the data matrix $\mathbf{Z}$ (Figure 7) after $\mathbf{A}^{\frac{1}{2}}$ scaling gives Figure 10, and then rotating and finally translating also gives Figure 8.

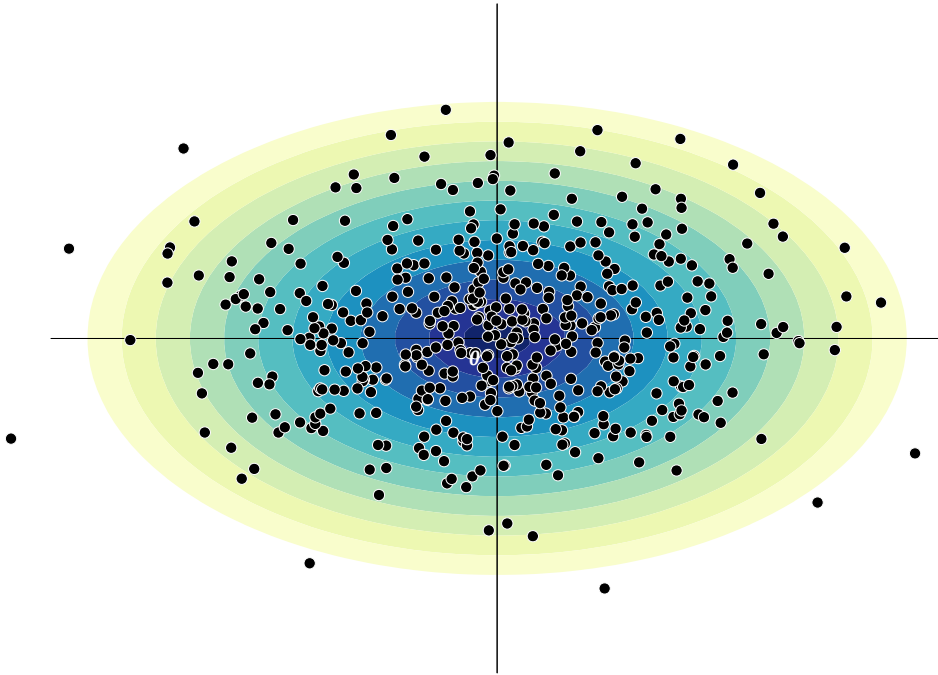


Figure 10. Data Matrix $\mathbf{Z} @ \mathbf{A}^{\frac{1}{2}}$

Comparing the two equations above, we find that they have the same form! Both are equivalent to taking the square root of $\boldsymbol{\Sigma}_X$.



Please use tools such as DeepSeek/ChatGPT to complete the following exercises for this section.

Q1. Please learn to use the `numpy.linalg.cholesky()` function to perform a Cholesky decomposition on the matrix introduced in this section.

<https://numpy.org/doc/2.2/reference/generated/numpy.linalg.cholesky.html>

Q2. Please learn to use the `scipy.linalg.ldl()` function to perform an LDL decomposition on the matrix introduced in this section.

<https://docs.scipy.org/doc/scipy/reference/generated/scipy.linalg.ldl.html>